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Alterations to the bronchial gene-expression effects of tobacco smoking in people living with HIV

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Abstract

Background: Although acquired immunodeficiency syndrome-related mortality has decreased since the introduction of antiretroviral therapy, the incidence of lung cancer in patients living with HIV (PLWH) remains high. Understanding the effects of tobacco smoke on airway gene expression in PLWH may provide insight into the elevated lung cancer risk associated with HIV.

Methods: Airway epithelial brushings were collected from 12 PLWH who smoke and 19 PLWH who have never smoked and profiled using RNA-seq. Linear models were used to identify genes differentially expressed between people who smoke and have never smoked and between people who previously smoked and have never smoked. Gene set enrichment analysis was used to validate smoking-related gene signatures across datasets. Data was integrated with gene expression data from a cohort of patients without HIV to identify genes differentially modulated by both smoking and HIV status.

Results: We found that previous smoking-related gene signatures are similarly associated with smoking in PLWH, including the persistent differential expression of a subset of “irreversibly altered” genes in people who previously smoked compared to people who have never smoked. We also identified and validated two gene signatures specifically modulated in people who smoke without HIV.

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Author Contributions

NHL and AA conceived the clinical study. NHL, AA, SR, and DK recruited the clinical cohort. MI performed the bronchoscopies for the clinical study. SR, DK, and AO assisted in bronchoscopies and sample collection. DK conducted the smoking cessation intervention. AO and RTP conducted laboratory analyses. ML performed data analyses. ML wrote the first draft of the manuscript. All authors critically reviewed the findings, contributed to the manuscript, and read and approved the final manuscript.

Conclusions: Many of the effects of smoking on bronchial gene expression are shared between PLWH and non-HIV individuals. Of the subset of genes that differ between PLWH and non-HIV individuals, it appears that the effect of smoking on gene expression is more attenuated in PLWH. This includes genes related to oxidative stress and epithelial integrity, and suggests potential mechanisms that may contribute to smoking-related outcomes that are more prevalent in PLWH, such as lung cancer.

Keywords

Tobacco smoke; smoking cessation; gene expression changes

Introduction

Highly active antiretroviral therapy (ART) has substantially extended the lifespan of people living with HIV (PLWH) and reduced AIDS-related morbidity and mortality¹. As this population ages, however, they face a rising burden of chronic diseases and non-AIDS-defining cancers (NADC's). PLWH experience disproportionately high rates of obstructive lung diseases, including chronic obstructive pulmonary disease (COPD) and asthma; an estimated 16–20% of PLWH have one of these diagnoses², and HIV infection itself has been identified as an independent risk factor for COPD³. Among NADC's, lung cancer is both the most common and the leading cause of cancer-related deaths in PLWH⁴. Compared to individuals without HIV, PLWH have higher incidence rates of lung cancer, are diagnosed at younger ages, and experience significantly lower 5-year survival rates^{5,6}.

Cigarette smoking, a well-established risk factor for lung cancer, is highly prevalent among PLWH, with rates exceeding those in the general population⁷. However, PLWH are at an elevated risk for lung cancer compared with HIV-uninfected individuals, even after controlling for smoking prevalence and intensity^{8,9}. Hypothesized mechanisms for this increased risk include HIV-associated inflammation¹⁰, lung damage from recurrent infections, which are more common in PLWH¹¹, or an HIV-mediated increase in susceptibility to tobacco carcinogens¹².

This phenomenon is not unique to HIV: immune dysregulation has emerged as a key contributor to lung cancer risk across diverse populations. Patients with rheumatoid arthritis have a more than 50% increased risk of lung cancer, even after accounting for smoking¹³. Similarly, solid organ transplant recipients have an approximately 1.5- to 6-fold higher lung cancer incidence compared to the general population¹⁴. These observations suggest that chronic immune dysfunction, whether due to autoimmune disease, immunosuppressive therapy, or HIV, may contribute to the development of lung cancer.

It has been previously demonstrated that gene-expression alterations in the normal-appearing airway epithelium of people who smoke and people who have previously smoked can serve as biomarkers of smoking-related lung diseases, such as lung cancer^{15,16} and COPD¹⁷. Cigarette smoke induces widespread molecular alterations in the airway epithelium, a “field of injury” characterized by dysregulation of genes involved in immune response, inflammation, and tissue repair¹⁸. Previous work has identified reversible and irreversible changes in gene expression in response to smoking in the bronchial epithelium¹⁹. However,

the extent to which these signatures may be modified by HIV has not been systematically investigated.

In this study, we analyzed bronchial epithelial gene expression from PLWH who smoke and have never smoked to characterize the smoking-associated changes and compare them to those observed in HIV-negative individuals. We then sought to identify gene expression effects of smoking that are modulated by HIV status, which may help uncover mechanisms underlying the heightened lung cancer risk in PLWH.

Methods

Study Population and Sample Collection

Individuals with HIV who smoke ($n = 12$) and have never smoked ($n = 19$) were recruited from Boston Medical Center (BMC) between June 2017 and October 2022 (NCT02836067). All participants were PLWH who had been on ART for at least 6 months and had documented viral suppression (<50 copies/mL) at the time of enrollment. Individuals with active malignancy, including lung cancer, or with COPD were excluded. The full inclusion and exclusion criteria for the study are listed in Supplemental Table 1. Bronchial brushings were obtained at baseline from all participants. Individuals who smoke participated in a 10-week smoking cessation program that included behavioral counseling and pharmacologic cessation support, primarily nicotine-replacement therapy (NRT), with varenicline also available. Follow-up bronchial brushings were obtained from 7 out of the 12 participants who smoked, with 5 of these individuals completing the cessation intervention, defined as maintaining abstinence for at least 10 weeks, verified through weekly exhaled carbon monoxide²⁰, cotinine^{21,22}, and anabasine testing²³. The study was reviewed and approved by the BMC institutional review board. All subjects provided written informed consent. Further information is provided in the Supplemental Methods.

Data preprocessing

RNA-seq data were processed using a standardized pipeline for read alignment, quantification, and quality assessment, with full details provided in the Supplementary Methods²⁴.

Identification of Smoking-Related Gene Expression Changes in PLWH

The *limma*²⁵ package (version 3.4.0) in R was used to identify differentially expressed genes between people who smoke and people who have never smoked at baseline while controlling for median transcript integrity number (TIN) and age. A linear model was then used to identify genes differentially expressed between people who have never smoked and follow-up samples from the participants who completed the smoking cessation program (i.e., people who have never smoked vs. people who previously smoked), with median TIN and age as covariates.

Beane et al. previously classified genes differentially expressed in people who smoke relative to people who have never smoked into those reversible or irreversible upon smoking cessation¹⁹. We assessed whether these gene sets were differentially expressed in our dataset

of PLWH using Gene Set Enrichment Analysis (GSEA)²⁶, where we pre-ranked genes by the t-statistic from the linear models described above.

Identification of Gene Expression Changes Associated with HIV and Smoking Status

To examine the potential interaction between smoking and HIV status, the baseline samples of the BMC cohort of PLWH were combined with samples from the Detection of Early Lung Cancer Among Military Personnel 2 (DECAMP-2)²⁷ study (NCT02504697). The DECAMP-2 cohort includes people who smoke and have previously smoked who are at high risk of developing lung cancer (e.g., history of COPD, emphysema, or at least one first-degree relative with a diagnosis of lung cancer) and are eligible for lung cancer screening by chest CT. We have previously reported bulk RNA-sequencing of mainstem bronchial brushes from DECAMP-2²⁸, and they are available from the Gene Expression Omnibus under accession GSE210659.

Before combining the two cohorts, we used a weighted-voting algorithm based on a previously described smoking-associated signature¹⁹ to predict the smoking status of the DECAMP-2 samples. For each sample in the BMC cohort, the DECAMP-2 patient with matched smoking status closest in age with the highest smoking status prediction strength was selected as the matched control. ComBat-Seq²⁹ was used to adjust for batch effects.

Genes whose smoking-associated differential expression is modulated by HIV were identified by incorporating an interaction term in *limma*, using TIN, age, and sex as covariates. Gene Set Variation Analysis (GSVA)³⁰ was used to summarize the expression of the interaction effect genes across samples.

Identification of Interaction Effect Gene Signature Biology

Biological processes and pathways enriched in the genes modulated by both HIV and smoking were identified using EnrichR³¹. Ensembl IDs were converted to Gene Symbols using biomaRt³², and the following databases were queried: Molecular Signatures Database (MSigDB), GO Biological Process 2017, KEGG 2019, WikiPathways 2016, Reactome 2016, and Biocarta 2016. Processes/pathways with an FDR < 0.05 and the number of overlapping genes > 1 were considered significantly enriched.

Validation Datasets

We evaluated the expression of the genes modulated by smoking and HIV status in an independent cohort reported by Hernández Cordero et al.³³ containing patients with and without HIV at St. Paul's Hospital in Vancouver. Residual count matrices adjusted for age, sex, COPD, and median TIN were used as input for signature scoring.

We also examined the expression of the interaction effect genes in three cohorts of patients without HIV: DECAMP-2, DECAMP-1^{27,28}, and de Biase et al.³⁴. DECAMP-1 (GSE210659, $n = 126$) consists of people who smoke or have previously smoked with indeterminate pulmonary nodules. De Biase et al. collected bronchial brushings from people who smoke, have previously smoked, and have never smoked, being investigated for suspicion of lung cancer. FastQ files were downloaded from the European Genome-

Phenome Archive³⁵ (EGAD50000000333) and processed similarly to the BMC cohort described above.

A ranked list was generated for each dataset comparing individuals who smoke and individuals who have previously smoked or have never smoked using the t-statistic. The enrichment of the interaction effect genes identified in the combined BMC / DECAMP-2 cohort was evaluated by GSEA.

Statistical Analysis

All statistical analyses were performed using R (4.4.0). ANOVA, t-tests, and chi-squared tests were used to analyze differences in baseline clinical variables between the participants who smoke and who have never smoked. False discovery rates for differential expression and enrichment analyses were calculated using the Benjamini-Hochberg method. For GSVA scores, the statistical significance of differences between groups was assessed using a t-test.

Results

Characteristics of the Study Population

Participants who smoke and have never smoked were well matched for demographic characteristics (Table 1). All participants had undetectable HIV-1 RNA levels (< 50 copies/mL). We detected no more than 54 reads per sample that aligned to the HIV genome. The average time between baseline and follow-up bronchoscopy was 18 weeks.

Rapidly reversible smoking genes are similarly changed in PLWH

Analysis of the BMC cohort of PLWH demonstrated that the gene signature previously identified by Beane et al.¹⁹ as rapidly reversible upon smoking cessation is concordantly differentially expressed between the individuals who smoke and have never smoked in this cohort. These reversible smoking genes were previously found to be enriched for genes involved in oxidoreductase activity and electron-transport activity, reflecting processes upregulated to metabolize the components present in cigarette smoke¹⁹. Unsupervised hierarchical clustering of the rapidly reversible smoking signature genes revealed strong co-expression of the signature genes and sample clustering that aligned with participant smoking status. The cluster of samples with A) elevated expression of the genes reversibly decreased in individuals who have previously smoked and B) decreased expression of the genes reversibly increased in individuals who previously smoked is enriched for individuals who currently smoke, while the other cluster is enriched for individuals who have never smoked (Chi-sq. $p < 1 \times 10^{-16}$; Figure 1A). Moreover, the longitudinal samples from study participants who quit smoking clustered with the samples from participants who have never smoked.

Next, we ranked genes from most increased to most decreased in PLWH who currently smoke relative to PLWH who have never smoked by the t-statistic from a linear model that adjusted for age, sequencing batch, and TIN. GSEA confirmed that the sets of genes Beane et al. found to be reversibly increased and decreased with smoking were significantly enriched in their respective directions in PLWH (Figure 1B).

Irreversible smoking-associated gene signature is conserved in PLWH

Beane et al. further identified genes in the bronchial epithelium that are altered in individuals who smoke and remain persistently altered in individuals who have previously smoked. We ranked genes from most increased to most decreased in PLWH who formerly smoked relative to PLWH who have never smoked by the t-statistic from a linear model adjusting for age, sequencing batch, and TIN. GSEA showed enrichment of Beane et al.'s persistently increased and decreased gene sets within this ranked list (Figure 2). These findings validate the persistently altered genes identified by Beane et al. and highlight that smoking has similar effects on the airway epithelium of people who have previously smoked, irrespective of HIV status.

HIV status alters smoking-associated gene expression patterns

Next, we sought to determine if any smoking-associated gene expression changes are modified by HIV status. To do so, we combined the baseline samples from the BMC cohort with age- and smoking- matched samples from DECAMP-2, a cohort of people not known to be infected with HIV. In the batch-corrected dataset of BMC and DECAMP-2 samples, we validated the Beane et al. reversible smoking signature (Supplemental Figure 1), confirming that batch correction did not abrogate known smoking-related gene expression patterns.

We then identified genes where the effect of smoking is modulated by HIV status using a linear model that included the main effects of smoking status and HIV status as well as a term for their interaction. This model also included age, sex, and median TIN as covariates. This model identified 51 genes with a smoking*HIV interaction effect p-value < 0.001 (Figure 3A, Supplemental Table 2). Based on our sample size, we estimate that our study is 70% powered to detect genes with a 2-fold smoking*HIV interaction effect (see Supplemental Methods), and as such, should be considered an exploratory analysis.

To evaluate whether duration of viral suppression influenced our findings, we adjusted for the duration of viral suppression in the subset of PLWH (n = 18) with available data on years since last detectable viral load (>50 copies/mL) and repeated the analysis using their matched HIV-negative counterparts. Using GSEA, the three gene clusters identified in the primary analysis were significantly enriched, indicating that adjustment for duration of viral suppression did not substantially affect the results (Supplemental Figure 2). We also failed to detect any genes with significant associations between gene expression and duration of viral suppression among PLWH (data not shown). These analyses suggest that the impact of duration of viral suppression on airway gene expression is minimal, and the observed interaction between smoking and HIV status is robust to variability in viral suppression duration.

Unsupervised hierarchical clustering divided the genes into three co-expression clusters (1–3). Enrichment scores for each gene cluster were calculated for each participant using Gene Set Variation Analysis (GSVA) (Figure 3B). Gene cluster 1 was increased in people who smoke compared to people who have never smoked in the participants without HIV but not affected by smoking in PLWH; gene cluster 2 was decreased in people who smoke

compared to people who have never smoked in participants without HIV but not affected by smoking in PLWH; gene cluster 3 was higher specifically in individuals who smoke with HIV.

Enrichment analysis identifies biological pathways linked to smoking-HIV interaction

To explore the molecular underpinnings of these gene expression patterns, we used enrichR to identify pathway annotations that are over-represented among the genes in the three clusters. Cluster 1 is enriched for genes involved in protein processing in the endoplasmic reticulum and genes encoding components of the apical junction complex, while cluster 2 was enriched for genes related to positive regulation of transcription from RNA polymerase II, as well as genes upregulated by KRAS signaling. Cluster 3 is enriched for the Gene Ontology biological process related to the regulation of autophagy (Supplemental Table 5).

HIV-smoking interaction gene signatures validate in independent cohorts

To test whether these interaction effect gene expression patterns were reproducible, we evaluated the three gene clusters in an independent bronchial epithelial dataset from Hernández Cordero et al.³⁶, which included³² PLWH with and without COPD and non-HIV adults with and without COPD. In this cohort, 88% of the PLWH were on antiretroviral therapy, and 76% had undetectable viral loads. Demographic information for this cohort can be found in Supplemental Table 3. To ensure the quality of this dataset, we assessed the enrichment of rapidly reversible smoking-related genes (Supplemental Figure 3) and found significant enrichment of both upregulated and downregulated genes.

GSVA was used to calculate a score for each gene cluster for each participant in the Hernández Cordero et al. cohort. A linear model was used to test if a gene cluster's GSVA score was significantly associated with the interaction term between smoking status (current vs. former/never) and HIV status. Of the three clusters, clusters 1 ($p = 0.013$) and 2 ($p = 0.026$) demonstrated a significant smoking*HIV interaction effect. We then compared the signature scores grouped by smoking status and confirmed that gene cluster 1 was specifically increased amongst individuals who smoke without HIV, while remaining unchanged in patients with HIV, and gene cluster 2 was significantly decreased amongst individuals who smoke without HIV, while remaining unchanged in patients with HIV (Figure 4).

We further assessed whether the smoking-associated expression pattern of clusters 1 and 2 could be replicated in three additional datasets of non-HIV patients: the remaining 207 samples from DECAMP-2 not included in our primary analysis; normal-appearing bronchial epithelium tissues collected from individuals with indeterminate pulmonary nodules participating in the DECAMP-1 study; and bronchial epithelium tissue from de Biase et al (Supplemental Table 4). In all three datasets, cluster 1 was significantly enriched among the genes whose expression increased in people who smoke, and cluster 2 was significantly enriched among the genes whose expression decreased in people who smoke by GSEA (Supplemental Figure 4A–C), demonstrating consistent replication across independent cohorts.

Discussion

The interplay between smoking, HIV infection, and lung cancer risk presents an opportunity to better understand how HIV modifies airway biology. Smoking is a well-established risk factor for lung cancer, COPD, and other pulmonary diseases, driven by its impact on airway epithelial cells, inflammation, and tissue repair pathways^{37,38}. In PLWH, the immune dysregulation caused by chronic HIV infection, even with suppressive ART, could alter the molecular response to smoking⁹. Exploring these interactions may enhance our understanding of smoking's role in elevating lung cancer risk in PLWH but also offer insight into lung cancer pathogenesis without HIV.

In this study, we demonstrated concordant gene expression changes of bronchial airway genes previously shown to be altered by smoking and revert to baseline upon cessation. This finding is consistent with Hijazi et al.³⁹, who reported that many of the genes induced by smoking in the airway epithelium return to baseline within 4 to 8 weeks after cessation in people without HIV, a timeframe consistent with the 10-week cessation program in the current study. We also saw altered expression of genes previously found to be persistently altered despite smoking cessation in the post-cessation samples from the current study, compared to samples from individuals who never smoked. Taken together, these results suggest a common smoking-associated airway field of reversible and persistent injury in individuals with and without HIV.

We also detected a subset of gene expression responses to smoking that differed between people with and without HIV. Specifically, two gene clusters showed altered expression only in individuals without HIV, suggesting that HIV is associated with an attenuated transcriptional response to cigarette smoke. Cluster 1, which was increased in HIV-negative individuals who smoke but unchanged in PLWH, is enriched for genes involved in endoplasmic reticulum (ER) stress responses and the apical junction. ER stress is a hallmark of smoking-related oxidative damage, triggering adaptive pathways to mitigate protein misfolding⁴⁰. Chronic HIV infection or ART-associated changes may limit the epithelial capacity to mount this adaptive response. Notably, cluster 1 included *CRB3*, a critical regulator of epithelial polarity and apical junction assembly implicated in the development of precancerous lesions⁴¹, raising the possibility that HIV infection alters junctional remodeling in response to smoking. Follow-up studies using air-liquid interface cultures or organoids derived from PLWH could assess whether cigarette smoke produces different barrier or polarity phenotypes compared to cells from HIV-negative individuals.

In contrast, the genes in cluster 2 were decreased exclusively in HIV-negative individuals who smoke. Several genes in this cluster have established associations with cancer or HIV biology, including *ETS1*, a transcription factor involved in HIV-1 replication^{42,43} and associated with tumor invasion in lung adenocarcinoma⁴⁴; Pleomorphic adenoma gene 1 (*PLAG1*), a proto-oncogene frequently amplified in lung⁴⁵ and salivary gland⁴⁶ cancer; and *PDGFC* (platelet-derived growth factor C), which is overexpressed in lung cancer⁴⁷ and associated with advanced tumor stage and poor prognosis⁴⁸. Together, findings from clusters 1 and 2 suggest that HIV may attenuate smoking-related responses related to ER stress, apical junction biology, and oncogenic signaling. These differences may reflect

HIV-associated immune or inflammatory modulation and warrant further investigation to understand how they contribute to heightened airway disease risk such as lung cancer and COPD in PLWH who smoke.

There are limitations to the present study. First, the sample size for our analyses validating the persistently altered smoking genes was small, given the few patients who was able to complete 10 weeks of cessation with the smoking cessation program. However, our findings suggest a robust trend warranting further investigation in larger cohorts. Second, due to the study population, we had to combine the BMC cohort with DECAMP-2 to identify genes modulated by both HIV and smoking status. DECAMP-2 is an older population compared to the smoking cessation cohort. Although our analyses were adjusted for age, we cannot exclude the possibility that the differences in the two cohorts may have affected our ability to accurately ascribe the interaction effect to the effect of HIV status. While we were able to validate our interaction effect clusters in the Hernández Cordero et al. cohort, it is important to note that PLWH were also younger than the HIV-negative participants in that study. Lastly, since all the patients in the BMC cohort and most patients in the Hernández Cordero et al. cohort were receiving ART, these results may not reflect gene expression changes observed in PLWH not on ART, whom we would expect to have heightened inflammation and abnormal immune function. Replication of these findings in additional cohorts is necessary to validate the observed associations and assess generalizability to other populations.

In conclusion, our study has demonstrated a largely similar gene expression response to smoking in patients with and without HIV. Specifically, we validated previously identified smoking-related gene signatures in PLWH, including genes that return to baseline after smoking cessation and those that remain persistently altered. We also identified and validated two clusters of genes that are either repressed or induced by smoking in HIV-negative individuals but were not affected by smoking in PLWH. Many of these genes play roles in pathways known to be involved in the response to smoking or have been implicated in lung or other cancers. Our hypothesis is that the failure to alter the expression of these genes in response to smoking in PLWH contributes to the elevated risk of lung cancer and diseases in these individuals. This altered response could accelerate smoking-related damage, impair repair mechanisms, or interact with HIV-driven inflammation and immune dysfunction to enhance lung cancer risk. These findings highlight the complex interplay between smoking, HIV, and ART and provide a foundation for future mechanistic studies to determine how these altered responses contribute to heightened airway disease vulnerability in PLWH who smoke. Improving our understanding of how HIV modulates the molecular impact of smoking could inform targeted smoking cessation interventions and lung cancer screening strategies for PLWH.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Potential Conflict of Interests

AA is an investigator for studies supported by Gilead Sciences, GlaxoSmithKline/ViiV Healthcare, Theratechnologies and DayZero Diagnostics and has participated in advisory boards for Gilead Sciences. NHL is an employee of Moderna Therapeutics and has received research funding from Gilead Sciences, GSK/ViiV, Merck, and DayZero Diagnostics. RTP, MEL, ML, XX, GL, DK, SR and AO declare no conflicts of interest.

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Data Availability

The sequencing reads from this study are available from dbGaP (pending), while the aligned sequence counts per gene and clinical data required to recreate the analysis are available from the Gene Expression Omnibus (GSE318533).

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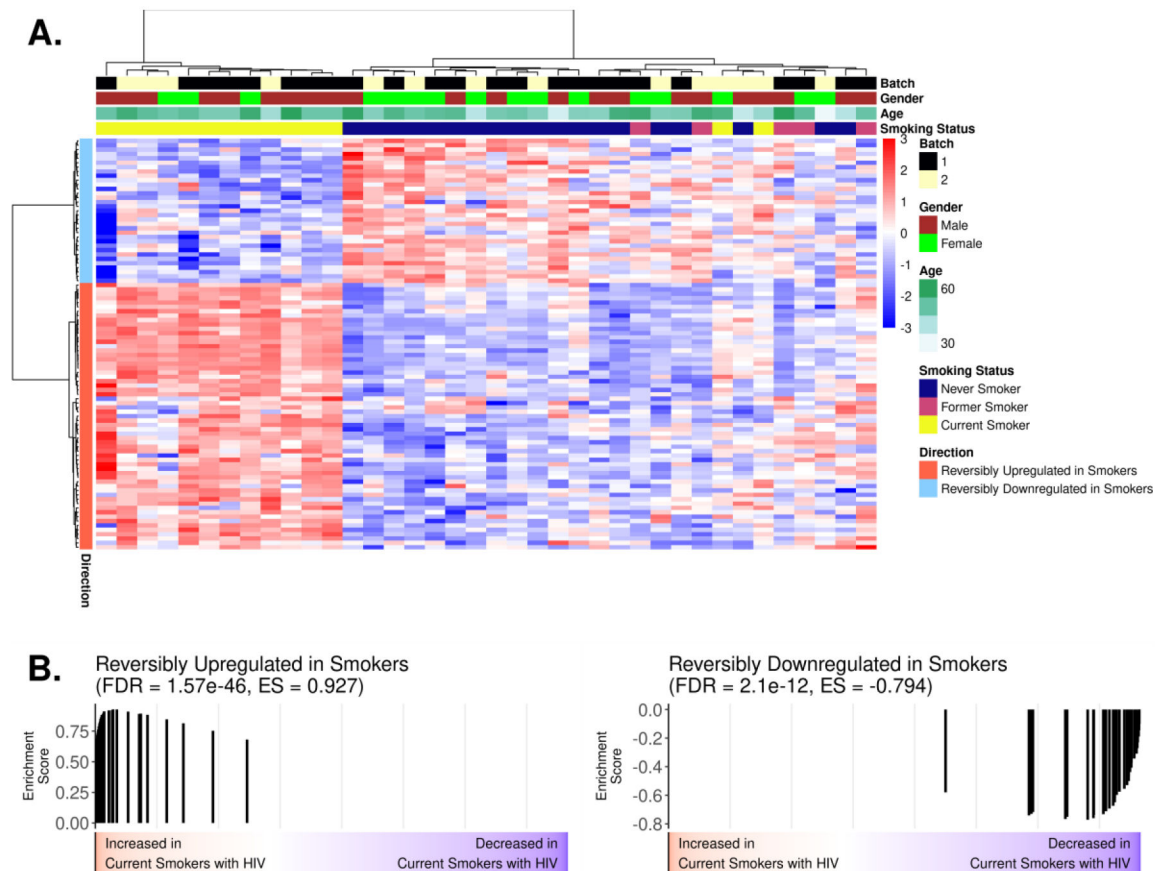


Figure 1: Conservation of the rapidly reversible smoking signature in PLWH.

(A) Heatmap shows the z-score normalized gene expression (residual of age and TIN) of the genes identified in Beane et al. 2007 as reversible upon smoking cessation. The Ward's minimum variance method was used to cluster the genes (rows) and patient samples (columns). There is a significant enrichment of participants who smoke in the left patient cluster compared to the right (χ^2 ; $p < 1 \times 10^{-16}$). (B) Similarities between the smoking-associated gene expression profiles found in Beane et al. and those found in the PLWH cohort were assessed using a pre-ranked gene set enrichment analysis (GSEA). Genes found by Beane et al. to have increased expression in individuals who smoke that are also reversible upon smoking cessation are enriched among the genes most induced in the bronchial epithelium of PLWH who smoke (left). The reverse is true for genes that were found by Beane et al. to have decreased expression in individuals who smoke that are also reversible upon smoking cessation (right). (ES = enrichment score)

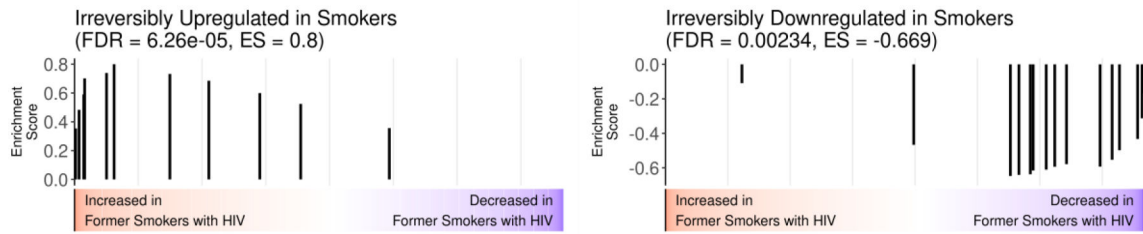


Figure 2: GSEA demonstrates enrichment of Beane et al. irreversible smoking signature.

Genes were ranked by t-statistic generated from a linear model between participants who quit smoking (n = 5) and participants who have never smoked (n = 19) in the PLWH cohort. Each vertical bar indicates the position of a gene whose expression in the bronchial epithelium is irreversibly altered by smoking within this ranked list. Both the genes persistently increased (left) and decreased (right) in people who previously smoked following smoking cessation were enriched.

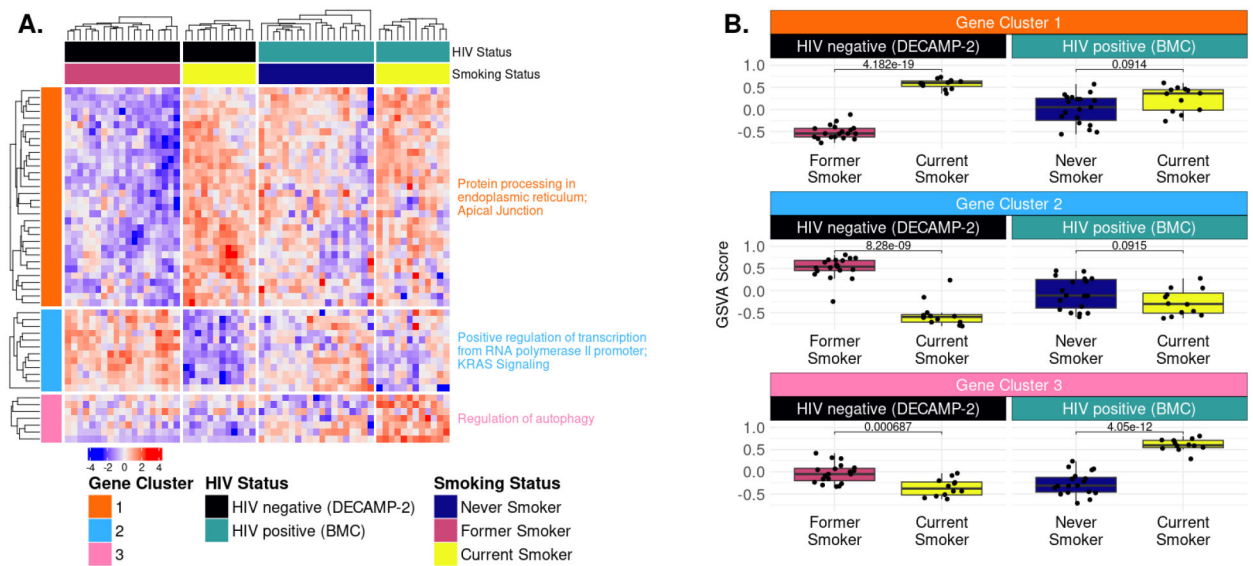


Figure 3: HIV-specific changes in gene expression induced by smoking

(A) Semi-supervised heatmap of z-score normalized gene expression after adjusting for median TIN for the 51 genes with a significant interaction effect between current smoking and HIV status ($p < 0.001$) in the 31 HIV negative samples (left) and 31 HIV positive samples (right). Based on hierarchical clustering, genes were grouped into three gene clusters (1–3). Biological pathways in which these clusters of genes were enriched in are shown on the side. Processes/pathways with an FDR < 0.05 and the number of overlapping genes > 1 were considered significantly enriched. (B). A signature score for each gene cluster was computed for each sample using Gene Set Variation Analysis (GSEA). The difference in signature score was compared by smoking status and HIV status via the t-test.

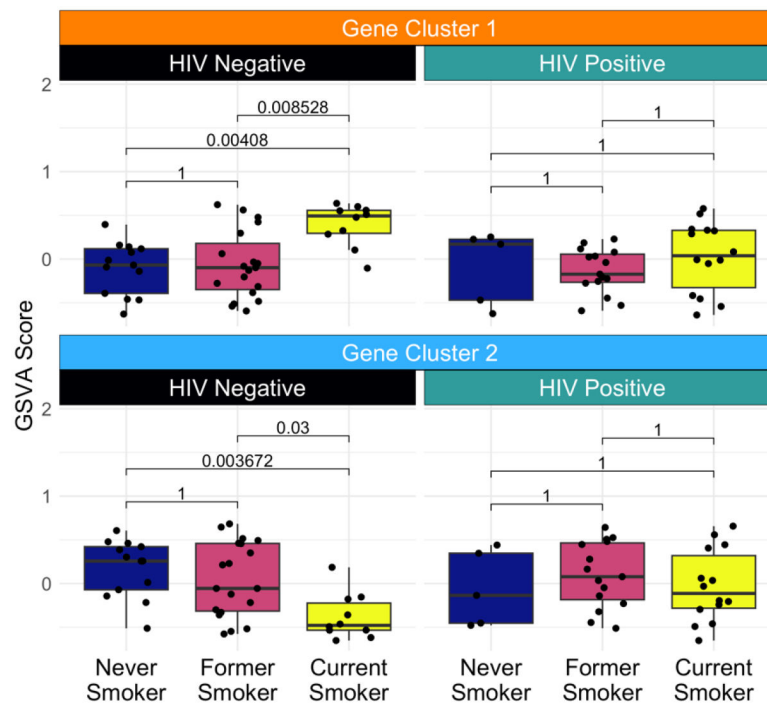


Figure 4. Gene Set Variation Analysis (GSVA) of gene clusters specific to individuals who smoke without HIV in an independent dataset.

GSVA was used to summarize the expression of gene clusters significant for the interaction effect between smoking and HIV status in the Hernandez Cordero dataset. The significance of differences in GSVA scores between groups was assessed via t-test.

Table 1:
Clinical and Demographic Characteristics for BMC Samples.

	Never Smoking (n = 19)	Current Smoking (n = 12)	p-value
Age (mean (SD))	47.1 (11.5)	54.7 (8.3)	0.058
Sex (%)			0.496
Male	9 (47.4)	8 (66.7)	
Ethnicity (%)			
Hispanic or Latino	2 (10.6)	2 (16.7)	1.0
Race (%)			0.689
White	3 (15.8)	4 (33.3)	
Black or African American	13 (68.4)	6 (50.0)	
Hispanic or Latino	1 (5.3)	1 (8.3)	
Not Available	2 (10.6)	1 (8.3)	
Smoking History (mean (SD))			
Year Smoking		31.5 (12.1)	
Pack Years		15.2 (8.0)	
Respiratory Diseases (%)			
Asthma	1 (5.3)	3 (25)	0.295
Obstructive Sleep Apnea	3 (15.8)	1 (8.3)	0.958
Chronic Sinusitis	1 (5.3)	0 (0)	1.0
Allergic Rhinitis	0 (0)	1 (8.3)	0.814
Inhaled Medication Use (%)	3 (15.8)	3 (25)	0.868
Median TIN (mean (SD))	72.4 (5.0)	74.5 (2.0)	0.179
CD4 Count (cells/mm3) (mean (SD))	672.3 (274.6)	821.8 (361.7)	0.233
>40 Reads aligned to HIV genome (%)	1 (5.2)	0 (0)	1.00
Years since last Viral Load >50 copies/ml			
Mean (SD)	2.7 (1.8)	3.1 (1.6)	0.59
Missing (%)	9 (47.4)	4 (33)	